



The Semester Project-V

The File System

Module Template

This article, which is part of the series on Linux device drivers, lays out a bare-bones file system module.

With the formatting of the pen drive, the file system is all set on the hardware. Now, it is time to decode that, using a corresponding file system module in kernel-space, and accordingly provide the user-space file-system interface for it to be browsed like any other file system.

The five sets of system calls

Unlike character or block drivers, file system drivers involve not just one but five structures of function pointers, for the various interfaces provided by a file system. These are:

- *struct file_system_type*: Contains functions to operate on the super block
- *struct super_operations*: Contains functions to operate on the inodes
- *struct inode_operations*: Has functions to operate on directory entries
- *struct file_operations*: Contains functions to operate on the file data (through page cache)
- *struct address_space_operations*: Has page cache operations for the file data

Along with these, came many new terms for Pugs.

He found the following glossary handy when it came to understanding various terms used in file system module development:

- *Page cache or buffer cache*: A pool of RAM buffers, each of *page* size (typically 4096 bytes). These buffers are used as the cache for the file data read from the underlying hardware, thus increasing the performance of file operations.

- *Inode*: A structure containing the meta-data or the information of a file, like permissions, the owner, etc. Though the file name is part of the file's meta-data, to conserve space, in typical Linux file systems it is not stored in the inode but in a directory entry. A collection of inodes is called an *inode table*.
- *Directory entry*: A structure containing the name and inode number of a file or directory. In typical Linux file systems, a collection of directory entries for the files and immediate sub-directories of, Directory D, for instance, is stored in the data blocks of the Directory D.
- *Superblock*: A structure containing information about the various data structures of the file system, like the inode tables, etc. Basically, it's meta-meta-data (meta-data for the meta-data).
- *Virtual File System (VFS)*: A conceptual file system layer that interfaces kernel-space to user-space in an abstract manner, showing 'everything' as a file, and which translates their operations from the user to the appropriate entity in the kernel space.

Each of the above five structures contains a list of function pointers, which need to be populated depending on what features are to be supported in the file system (module). For example, *struct file_system_type* may contain system calls for mounting and unmounting a file system—basically operating on its super block; *super_operations* may contain inode read/write system calls; *inode_operations* may contain a function to look up directory entries; and *file_operations* may generically operate on the page-cached file data, which may in turn invoke page